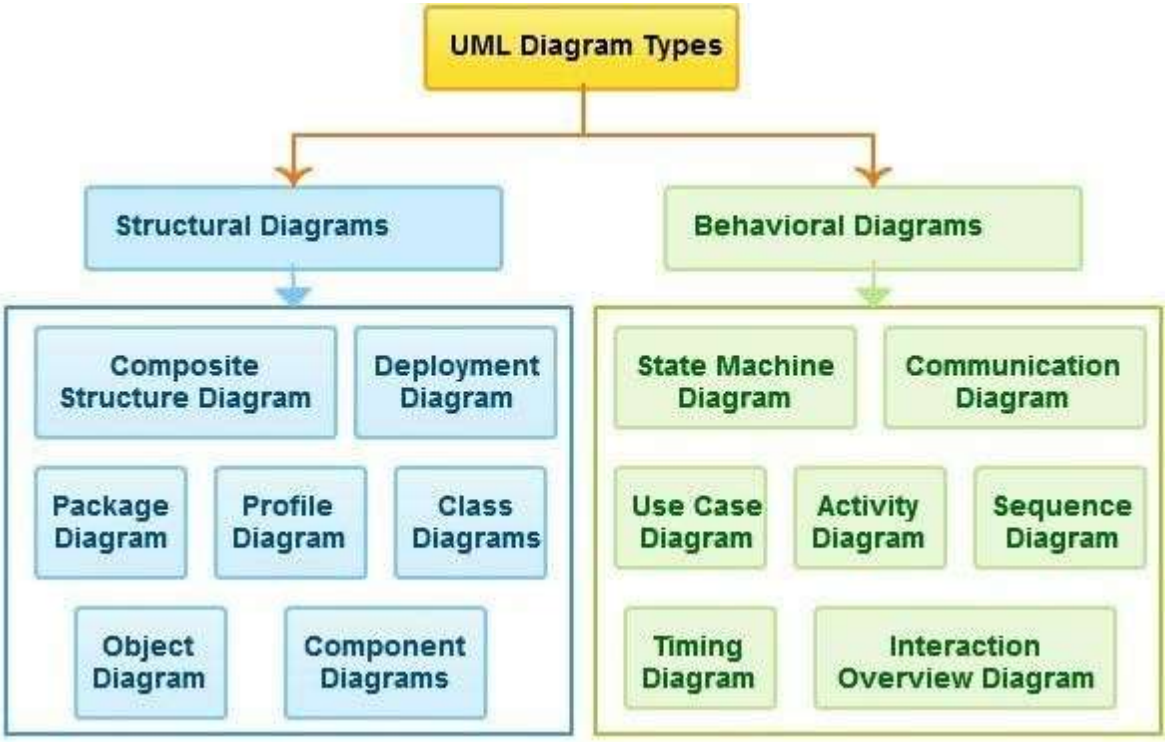
Introduction to UML

- **UML** – Unified Modeling Language diagram is designed to let developers and customers view a software system from a different perspective and in varying degrees of abstraction.
- One reason UML has become a standard modeling language is that it is programming-language independent.
- Since UML is not a methodology, it does not require any formal work products.
- In an effort to promote Object Oriented designs, three leading object oriented programming researchers joined ranks to combine their languages:
    - **i.** Grady Booch (BOOCH)
    - **ii.** James Rumbaugh (OML: object modeling technique)
    - **iii.** Ivar Jacobsen (OOSE: object oriented software eng) and come up with an industry standard [mid 1990's].

Types of UML Diagram



Structural Diagrams

- Structure diagrams emphasize on the things that must be present in the system being modeled.
- Since structure diagrams represent the structure, they are used extensively in documenting the software architecture of software systems.

Behavioural Diagrams

- Behavior diagrams emphasize on what must happen in the system being modeled.
- Since behavior diagrams illustrate the behavior of a system, they are used extensively to describe the functionality of software systems.

Use Case Diagram

Introduction:

- A use case diagram describes how a system interacts with outside actors.
- It is a graphical representation of the interaction among the elements and system.
- Each use case representation a piece of functionality that a system provides to its user.
- Use case identifies the functionality of a system.
- Use case diagram allows for the specification of higher level user goals that the system must carry out.
- These goals are not necessarily to tasks or actions, but can be more general required functionality of the system.
- You can apply use case to capture the intended behavior of the system you are developing, without having to specify how that behavior is implemented.
- A use case diagram at its simplest is a representation of a user's interaction with the system and depicting the specifications of a use case.
- A use case diagram contains four components.
  i. The boundary, which defines the system of interest in relation to the world around it.
  ii. The actors, usually individuals involved with the system defined according to their roles.
  iii. The use cases, which the specific roles are played by the actors within and around the system.
  iv. The relationships between and among the actors and the use cases.

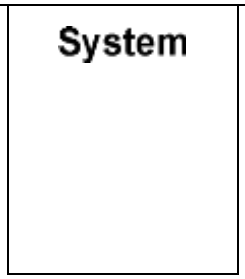
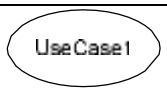
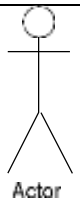
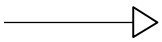
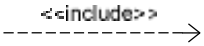
Purpose:

- The main purpose of the use case diagram is to capture the dynamic aspect of a system.
- Use case diagram shows, what software is suppose to do from user point of view.
- It describes the behavior of system from user's point.
- It provides functional description of system and its major processes.
- Use case diagram defines the scope of the system you are building.

When to Use: Use Cases Diagrams

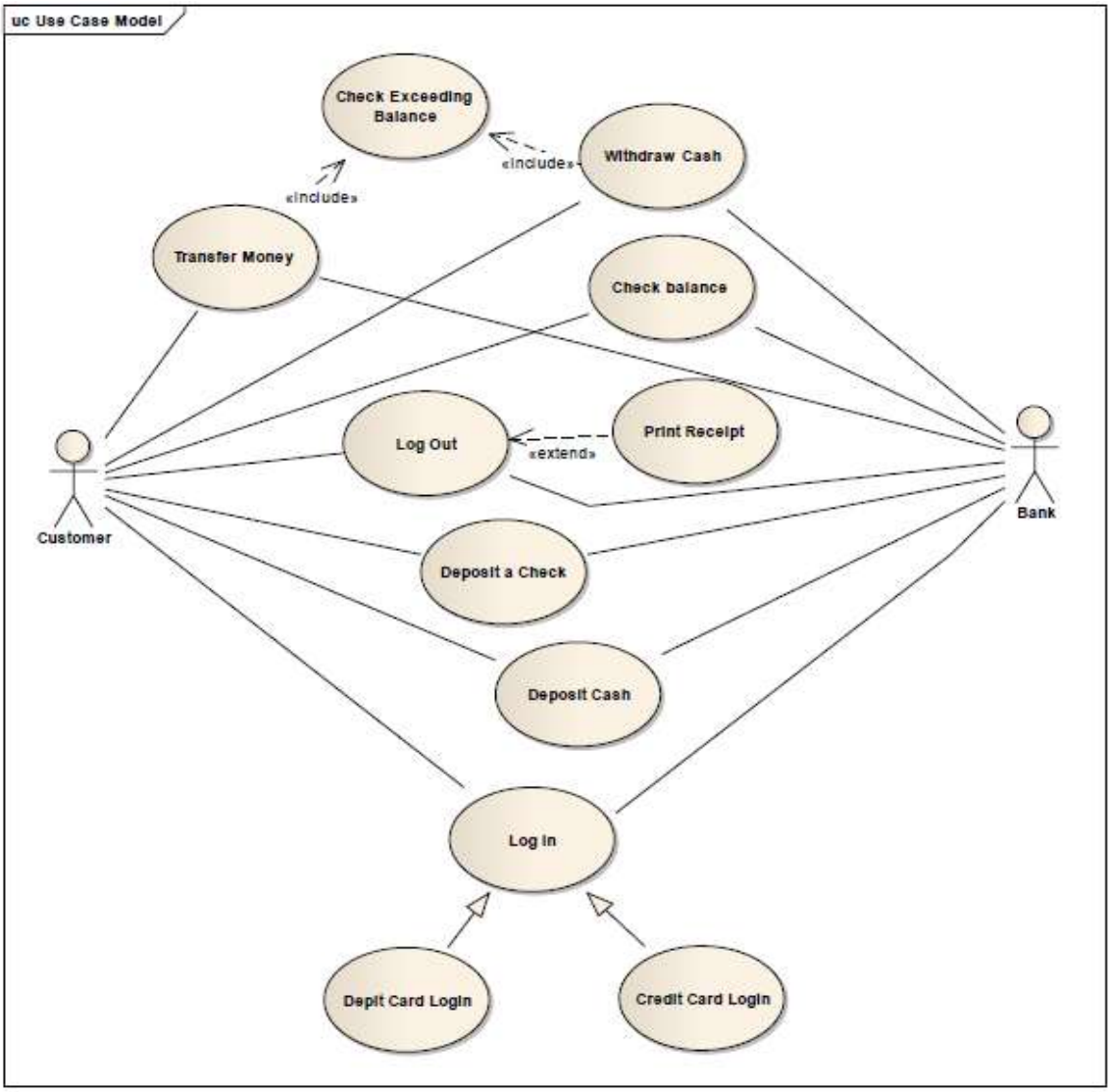
- Use cases are used in almost every project.
- They are helpful in exposing requirements and planning the project.
- During the initial stage of a project most use cases should be defined.

Use Case Notations

No.	Name	Notation	Description
1	System boundary		The scope of a system can be represented by a system boundary. The use cases of the system are placed inside the system boundary, while the actors who interact with the system are put outside the system. The use cases in the system make up the total requirements of the system.
2	Use case		A use case represents a user goal that can be achieved by accessing the system or software application.
3	Actor		Actors are the entities that interact with a system. Although in most cases, actors are used to represent the users of system, actors can actually be anything that needs to exchange information with the system. So an actor may be people, computer hardware, other systems, etc. Note that actor represent a role that a user can play, but not a specific user.
4	Association		Actor and use case can be associated to indicate that the actor participates in that use case. Therefore, an association corresponds to a sequence of actions between the actor and use case in achieving the use case.
5	Generalization		A generalization relationship is used to represent inheritance relationship between model elements of same type.
6	Include		An include relationship specifies how the behavior for the inclusion use case is inserted into the behavior defined for the base use case.

7	Extends	<<Extends>>	An extend relationship specifies how the behavior of the extension use case can be inserted into the behavior defined for the base use case.
8	Constraint	{condition}	Show condition exists between actors an activity.
9	Package	Package	Package is defined as collection of classes. Classes are unified together using a package.
10	Interface	Interface	Interface is used to connect package and use-case. Head is linked with package and tail linked with use-case.
11	Note		Note is generally used to write comment in use-case diagram.
12	Anchor		Anchor is used to connect a note the use case in use case diagram

Use-case Diagram For ATM System



Steps:

Step 1: First an Actor is Created and named as User/Customer.
Step 2: Secondly a system is created for ATM.
Step 3: A use case Enter PIN, Withdraw money is created and connected with user as association relationship.
Step 4: Similarly various use cases like Deposit money, Balance Enquiry, Manage Account etc are created and appropriate relationships are associated with each of them.

Class Diagram

Introduction

- The class diagram is a static diagram.
- A class model captures the static structure of a system by characterizing the objects in the system, the relationship between the objects, and the attributes and operations for each class of objects.
- The class diagram can be mapped directly with object oriented languages.
- The class model is the most important amonethe three models.
- Class diagram provide a graphical notation for modeling classes and their relationship.
- They are concise, easy to understand, and work well in practice.
- Class diagrams are the backbone of almost every object-oriented method including UML.
- They describe the static structure of a system.

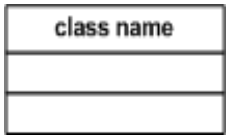
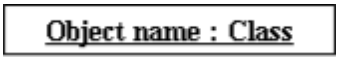
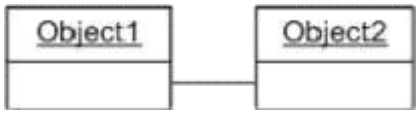
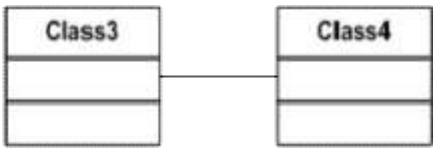
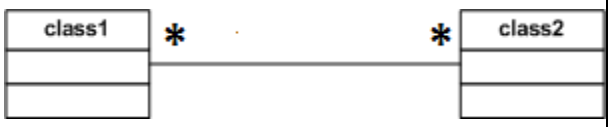
Purpose

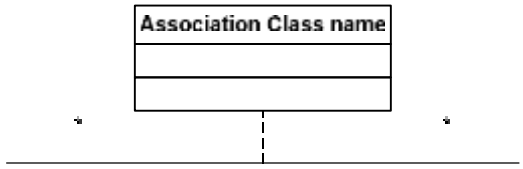
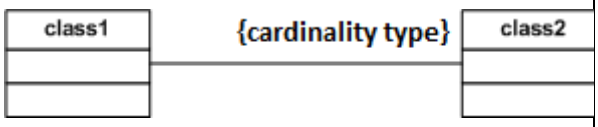
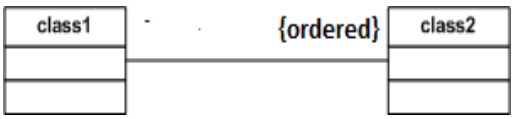
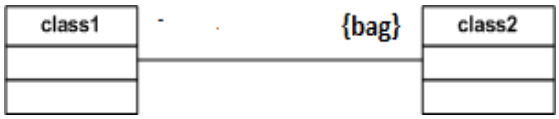
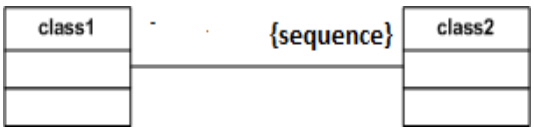
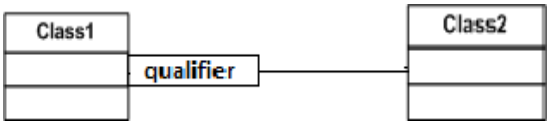
- Analysis and design of the static view of an application.
- Describe responsibilities of a system.
- Base for component and deployment diagrams.

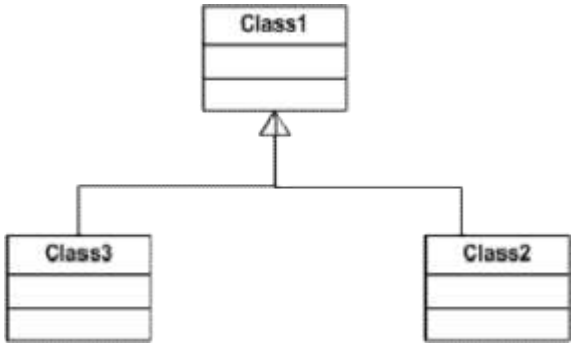
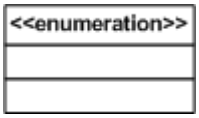
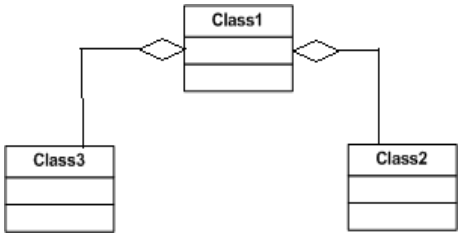
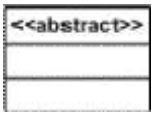
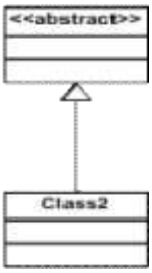
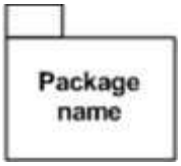
When to use : Class Diagram

- Useful for Forward and Reverse engineering.
- Class diagrams are useful both for abstract modeling and for designing actual programs.
- Developer uses class diagram for implementation decision.
- Business analysts can use class diagrams to model systems from the business perspective.

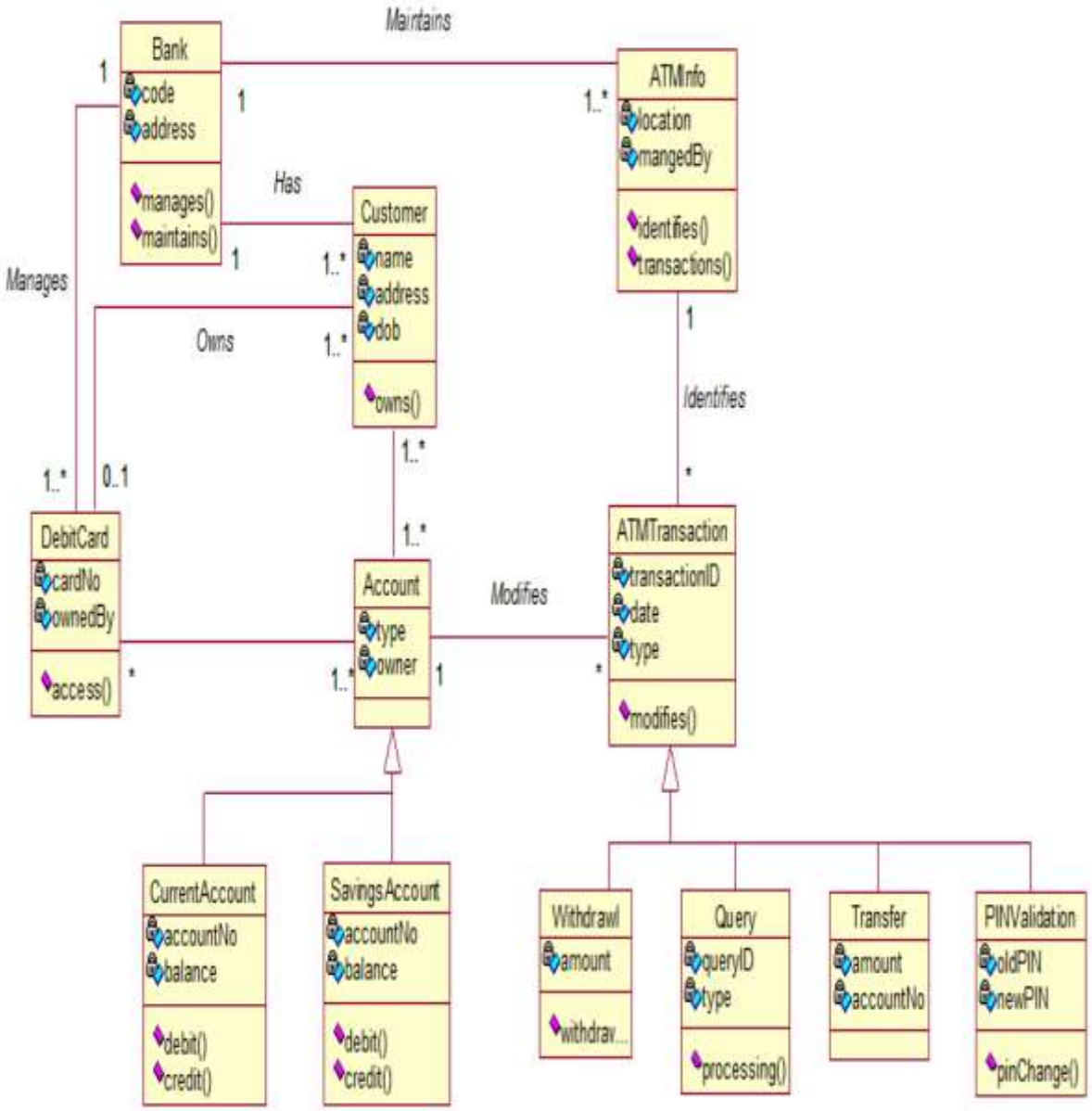
Class Diagram Notations

Sr. No.	Name	Symbol	Meaning
1.	Class		Class is an entity of the class diagram. It describes a group of objects with same properties & behavior.
2.	Object		An object is an instance or occurrence of a class.
3.	Link		A link is a physical or conceptual connection among objects
4.	Association		An association is a description of a links with common structure & common semantics.
5.	Multiplicity	Ex. 1 to 1 1 to * * to * * to 1 1 to 0....2 	Multiplicity specifies the number of instances of one class that may relate to a single instance of an associated class. It is a constraint on the cardinality of a set.

6.	Association class		It is an association that is a class which describes the association with attributes.
7.	cardinality		It describes the count of elements from collection.
8.	ordering		It is used to indicate an ordered set of objects with no duplication allowed.
9.	bag		A bag is a collection of unordered elements with duplicates allowed.
10.	sequence		A sequence is an ordered collection of elements with duplicates allowed.
11.	qualified association		Qualification increases the precision of a model. It is used to avoid many to many multiplicities and it converts into one to one multiplicity.

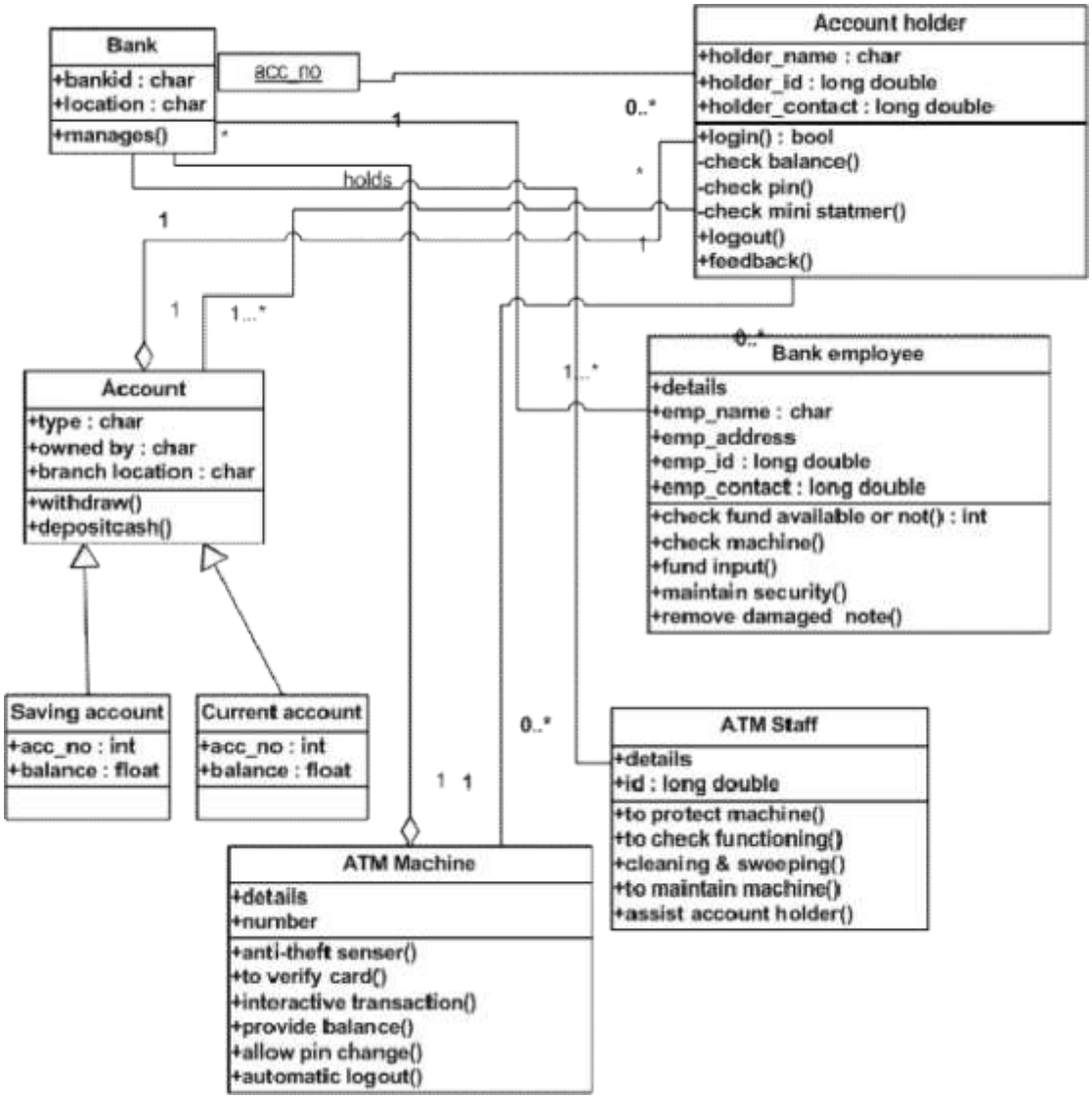
12.	generalization		Generalization organizes classes by their super-class and sub-class relationship.
13.	enumeration		An enumeration is a data type that has a finite set of values.
14.	aggregation		It is a strong form of association in which an aggregate object is made of constituent parts.
15.	composition		It is a form of aggregation. Composition implies ownership of the parts by the whole.
16.	Abstract class		It is a class that has no direct instances.
17.	Concrete class		It is a class that is intangible; it can have direct instances. Class-2 is example of concrete class
18.	package		A package is a group of elements with common theme.

Class Diagram for ATM



Step 1: First Classes are created.
Step 2: Named as PinValid, Account Type, Transaction, Update, Server, Customer classes are created.
Step 3: Appropriate relationships are provided between them as association.

Class Diagram for ATM



State Diagram

Introduction

- *A **state diagram** is a graph in which nodes correspond to states and directed arcs correspond to transitions labeled with event names.*
- A state diagram combines states and events in the form of a network to model all possible object states during its life cycle, helping to visualize how an object responds to different stimuli.
- A state diagram is a graph whose nodes are states and whose directed arcs are transitions between states.
- A state diagram specifies the state sequence caused by event sequence.
- State names must be unique within the scope of a state diagram.
- All objects in a class execute the state diagram for that class, which models their common behavior.
- We can implement state diagrams by direct interpretation or by converting the semantics into equivalent programming code.

Purpose

- The state model describes those aspects of objects concerned with time and the sequencing of operations events that mark changes, states that define the context for events, and the organization of events and states.
- They are used to give an abstract description of the behavior of a system.
- It provides direction and guidance to the individual counties within the states.
- It specifies the possible states, what transitions are allowed between states.
- It describes the common behavior for the objects in a class and each object changes its behavior from one state to another.
- It is used to describe the dependence of the functionality on the state of the system that is how the functionality of an object depends on its state and how its state changes as a result of the events that it receives.
- It describes dynamic behavior of the objects of the system.

When to use: State Diagram

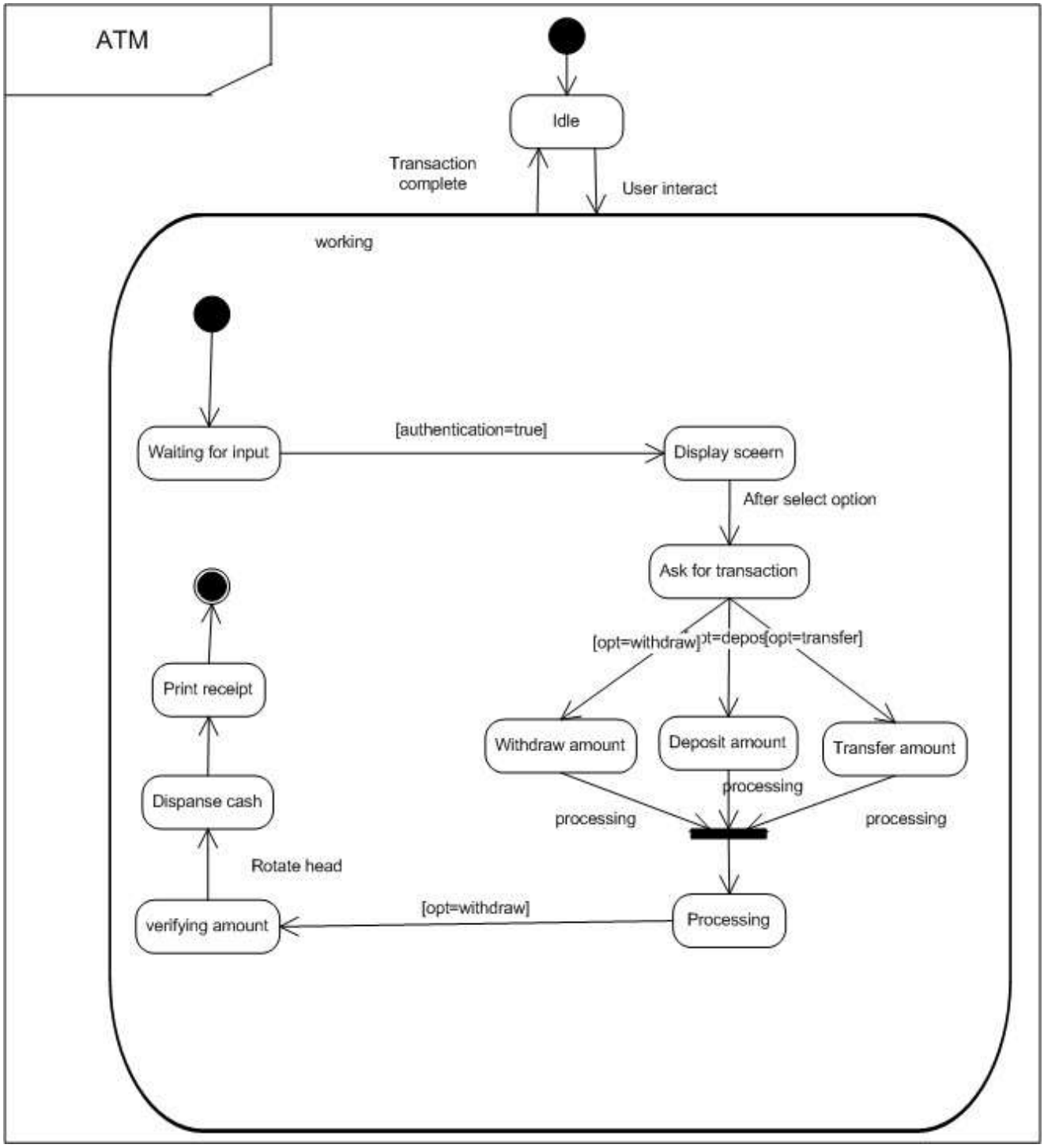
- They are perfectly useful to model behavior in real time system.
- Each state represents a named condition during the life of an object during which it satisfies some condition or waits for some event.
- It determines how objects of that class react to events.
- For each object state, it determines what actions the object will perform when it receives an event.

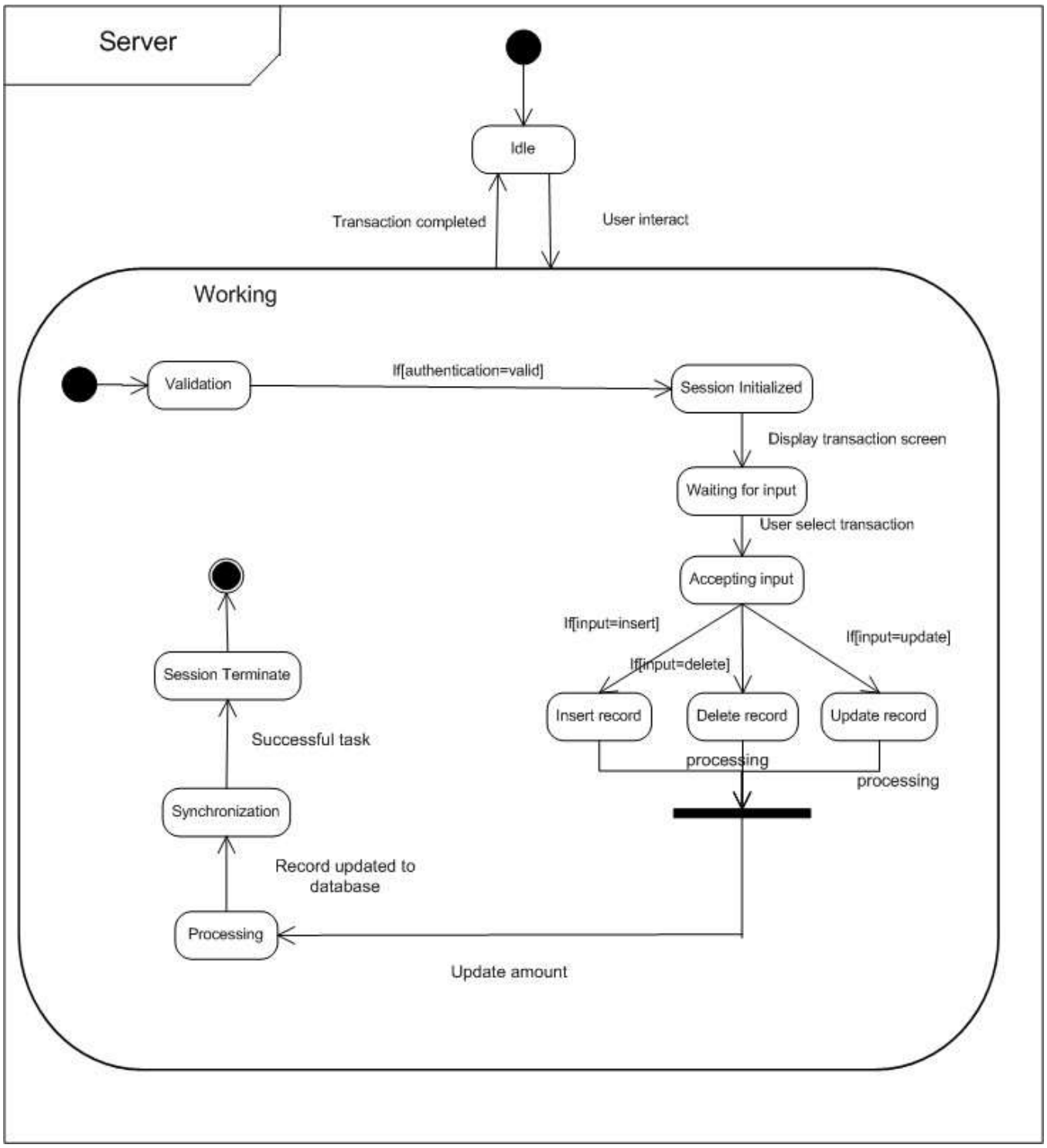
State Diagram Notations

No.	Name	Notation	Description
1	State	State	A state is an abstraction of the values and links of an object. State models a situation during which some (usually implicit) invariant condition holds.
2	Transition		A transition is a directed relationship between a source state and a target state. It may be part of a compound transition, which takes the state machine from one state configuration to another
3	Event	Event	A transition is an instantaneous change from one to another state
4	Change Event	When (Condition)	A change in value of a Boolean expression
5	Time Event	at (time condition)	The arrival of an absolute time or the passage of a relative amount of time
6	Signal Event	<<signal>> Collision force:Float	Receipt of an explicit, named, asynchronous communication among objects.
7	Guarded transition	[guard condition]	A guard condition is a Boolean expression that must be true in order for a transition to occur.
8	Do activity	do / Activity	A do activity an activity that continuous for extended time within state.

9	Entry activity	Entry / Activity	An state is entered by any incoming transition the entry activity is performed
10	Exit activity	Exit / Activity	When the state is exited by any outgoing transition the exit activity is performed
11	Nested State Diagram Sub machine Diagram	Local state name : sub state name	A submachine state specifies the insertion of the specification of a submachine. The state machine that contains the submachine state is called the containing state machine.
12	Composite State	CompositeState InternalStateA InternalStateB	A state can be refined hierarchically by composite states.
13	Activity effect	Event / effect	An activity is actual behavior that can be invoked by any number of effects
14	Initial state point		It shows the starting state of object.
15	Final state point		It shows the terminating state of object.

State diagram for ATM Management System





State diagram for ATM Management System

Sequence Diagram

Sequence Diagram

The sequence diagram represents the flow of messages in the system and is also termed as an event diagram. It helps in envisioning several dynamic scenarios. It portrays the communication between any two lifelines as a time-ordered sequence of events, such that these lifelines took part at the run time. In UML, the lifeline is represented by a vertical bar, whereas the message flow is represented by a vertical dotted line that extends across the bottom of the page. It incorporates the iterations as well as branching.

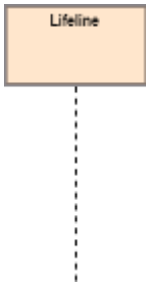
Purpose of a Sequence Diagram

- To model high-level interaction among active objects within a system.
- To model interaction among objects inside a collaboration realizing a use case.
- It either models' generic interactions or some certain instances of interaction.

Notations of a Sequence Diagram

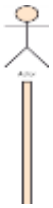
Lifeline

An individual participant in the sequence diagram is represented by a lifeline. It is positioned at the top of the diagram.



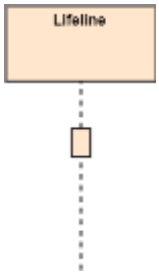
Actor

A role played by an entity that interacts with the subject is called as an actor. It is out of the scope of the system. It represents the role, which involves human users and external hardware or subjects. An actor may or may not represent a physical entity, but it purely depicts the role of an entity. Several distinct roles can be played by an actor or vice versa.



Activation

It is represented by a thin rectangle on the lifeline. It describes that time period in which an operation is performed by an element, such that the top and the bottom of the rectangle is associated with the initiation and the completion time, each respectively.

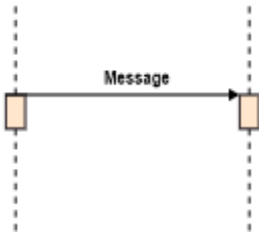


Messages

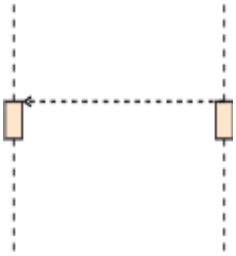
The messages depict the interaction between the objects and are represented by arrows. They are in the sequential order on the lifeline. The core of the sequence diagram is formed by messages and lifelines.

Following are types of messages enlisted below:

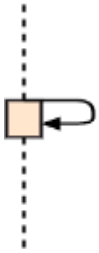
Call Message: It defines a particular communication between the lifelines of an interaction, which represents that the target lifeline has invoked an operation.



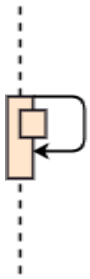
- **Return Message:** It defines a particular communication between the lifelines of interaction that represent the flow of information from the receiver of the corresponding caller message.



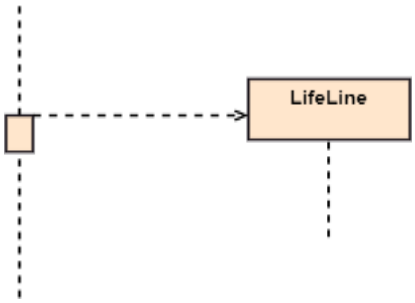
- **Self-Message:** It describes a communication, particularly between the lifelines of an interaction that represents a message of the same lifeline, has been invoked.



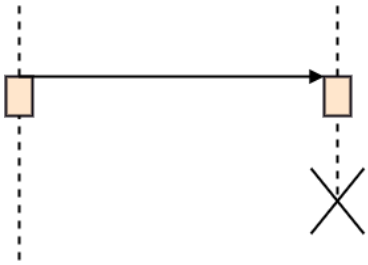
- **Recursive Message:** A self-message sent for recursive purpose is called a recursive message. In other words, it can be said that the recursive message is a special case of the self-message as it represents the recursive calls.



- **Create Message:** It describes a communication, particularly between the lifelines of an interaction describing that the target (lifeline) has been instantiated.



- **Destroy Message:** It describes a communication, particularly between the lifelines of an interaction that depicts a request to destroy the lifecycle of the target.



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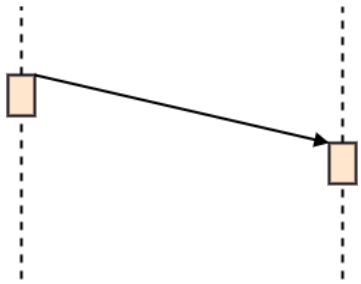
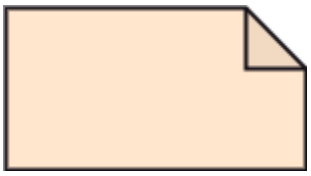
- **Duration Message:** It describes a communication particularly between the lifelines of an interaction, which portrays the time passage of the message while modeling a system.

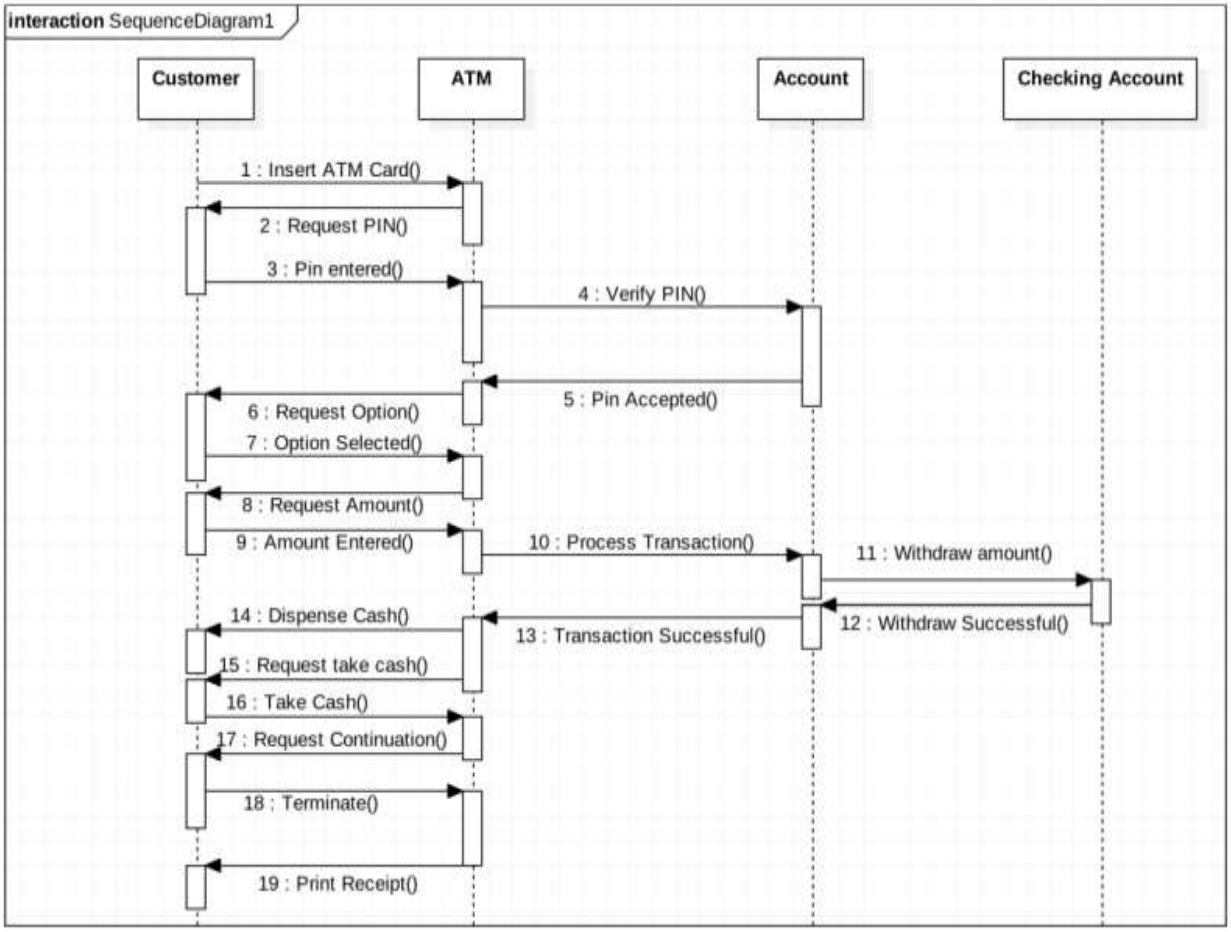


Note

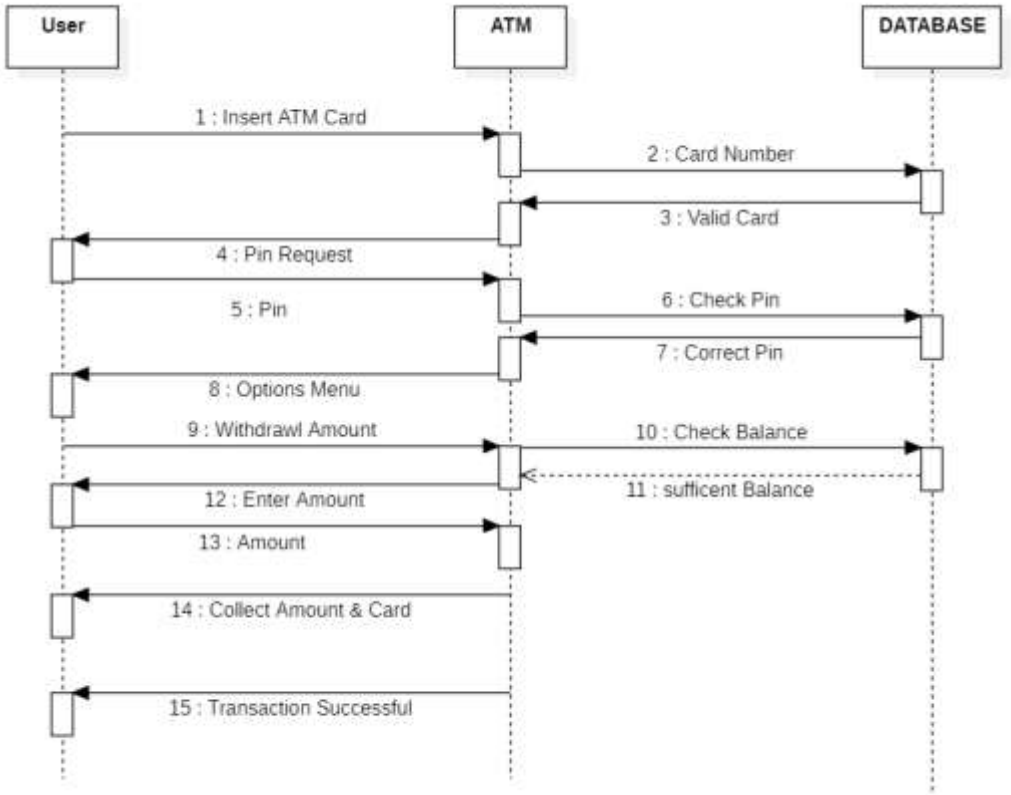
A note is the capability of attaching several remarks to the element. It basically carries useful information for the modelers.



Example of Sequence diagram for withdrawing amount from ATM

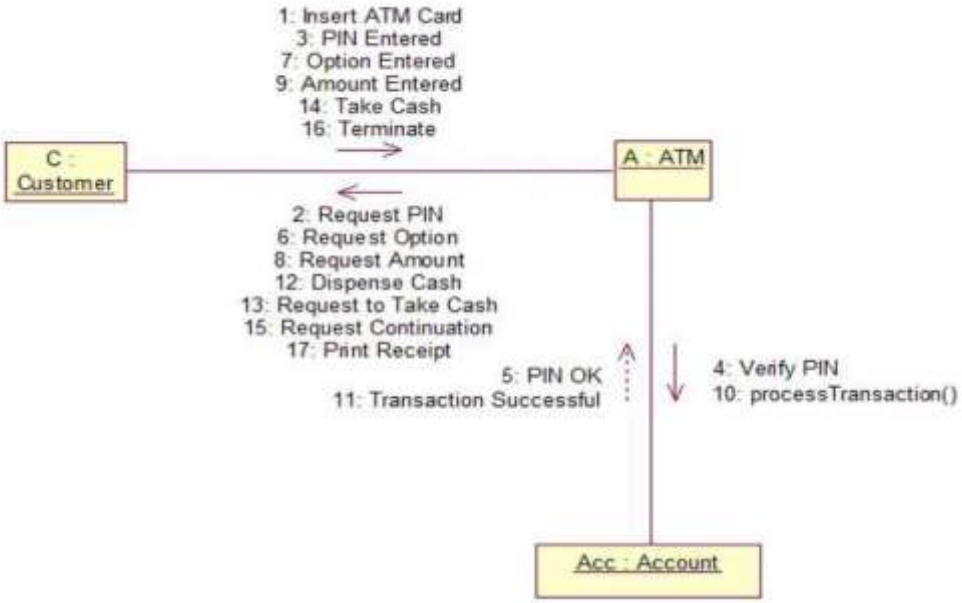


Step 1: First An actor is created and named as user.
Step 2: Secondly an object is created for Atm.
Step 3: Timelines and lifelines are created automatically for them.
Step 4: In sequence diagram interaction is done through time ordering of messages. So appropriate messages are passed between user and ATM


User
ATM
DATABASE
1 : Insert ATM Card
2 : Card Number
3 : Valid Card
4 : Pin Request
5 : Pin
6 : Check Pin
7 : Correct Pin
8 : Options Menu
9 : Withdrawl Amount
10 : Check Balance
11 : sufficent Balance
12 : Enter Amount
13 : Amount
14 : Collect Amount & Card
15 : Transaction Successful

Communication/Collaboration Diagrams

A Communication or Collaboration diagram, as shown is a directed graph that uses objects and actors as graph nodes. The focus of the collaboration diagram is on the roles of the objects as they interact to realize a system function. Directional links are used to indicate communication between objects. These links are labeled using appropriate messages. Each message is prefixed with a sequence number indicating the time ordering needed to realize the system function.



Step 1: First an actor is created and named as user.
Step 2: Secondly an object is created for ATM.
Step 3: In collaboration diagram interaction is done through organization.
Step 4: So appropriate messages are passed between user and ATM

Activity Diagram

Introduction

- An activity diagram is a type of flow chart with additional support for parallel behavior.
- This diagram explains overall flow of control.
- Activity diagram is another important diagram in UML to describe dynamic aspects of the system.
- Activity diagram is basically a flow chart to represent the flow from one activity to another activity
- The activity can be described as an operation of the system.
- The control flow is drawn from one operation to another. This flow can be sequential, branched or concurrent. This distinction is important for a distributed system.
- Activity diagrams deals with all type of flow control by using different elements like fork, join etc.

Purpose

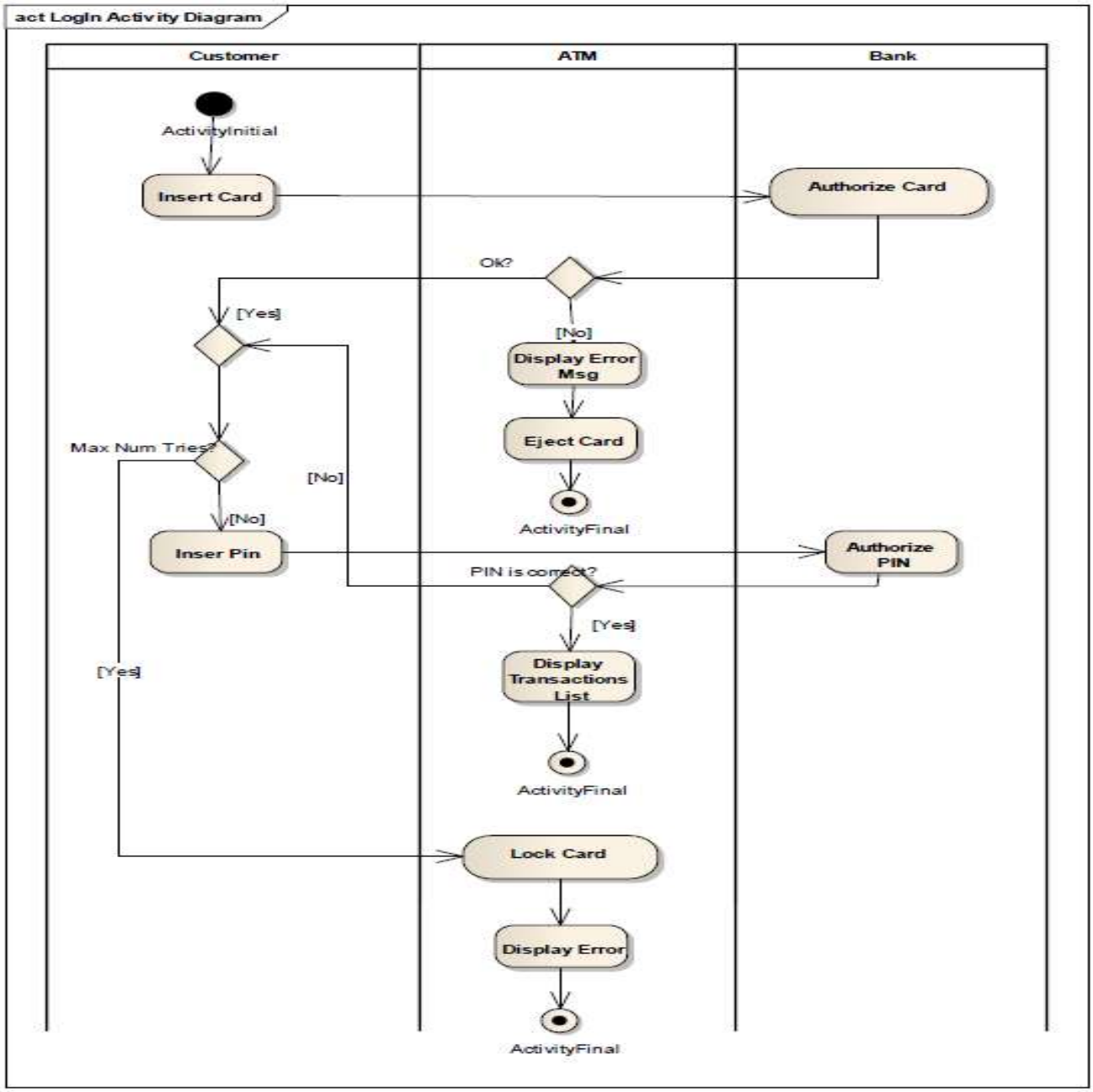
- Contrary to use case diagrams, in activity diagrams it is obvious whether actors can perform business use cases together or independently from one another.
- Activity diagrams allow you to think functionally.

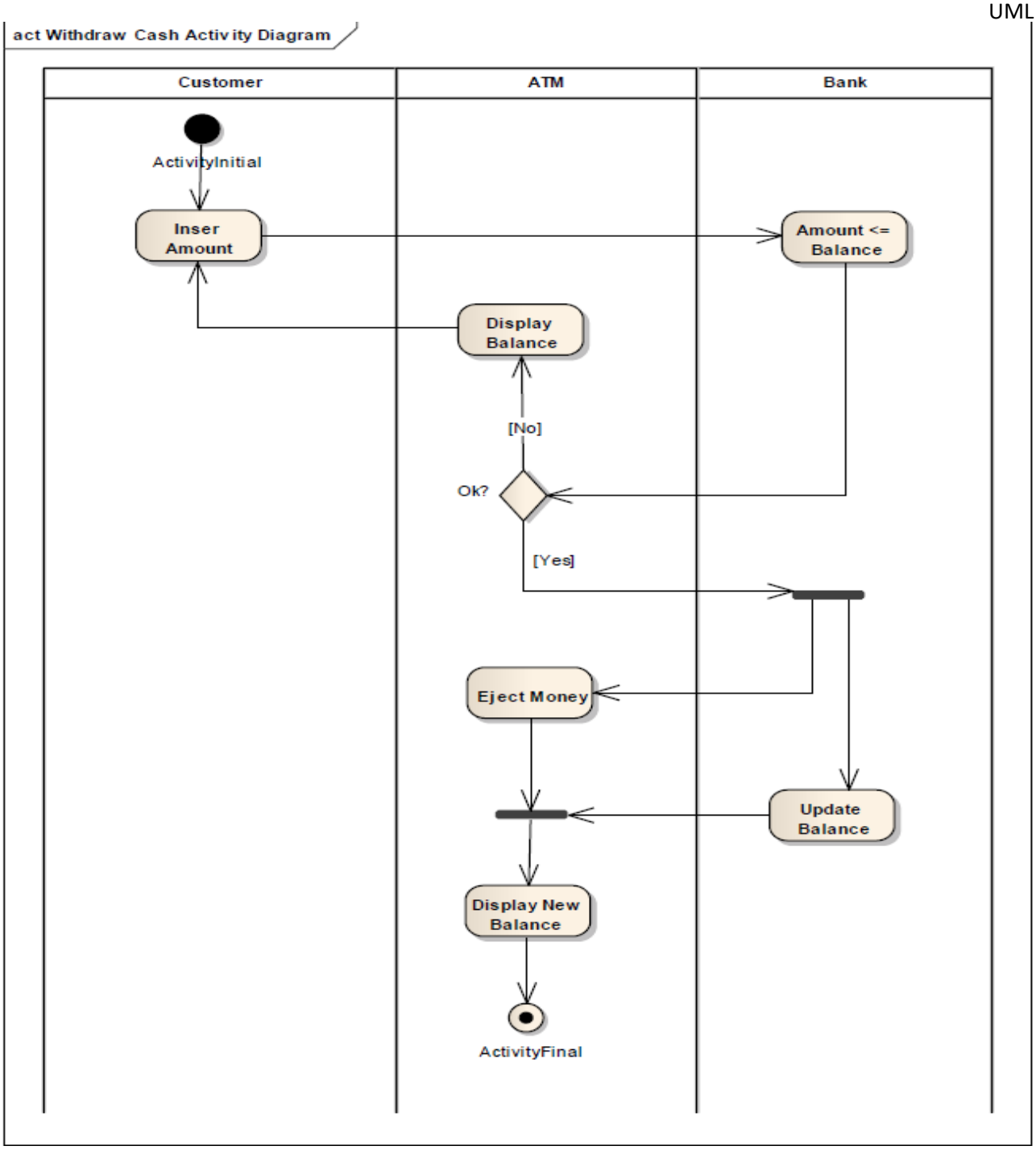
When to use : Activity Diagrams

- Activity diagrams are most useful when modeling the parallel behavior of a multithreaded system or when documenting the logic of a business process.
- Because it is possible to explicitly describe parallel events, the activity diagram is well suited for the illustration of business processes, since business processes rarely occur in a linear manner and often exhibit parallelisms.
- This diagram is useful to investigate business requirements at a later stage.
- An activity diagram is drawn from a very high level. So it gives high level view of a system. This high level view is mainly for business users or any other person who is not a technical person.
- This diagram is used to model the activities which are nothing but business requirements.
- So the diagram has more impact on business understanding rather *implementation details*.

Activity Diagram Notations

No.	Name	Symbol	Description
1.	Activity	Activity	Represent individual activity of system.
2.	Transition		Represents flow of data from one activity to another.
3.	Decision		Decision node is a control node that accepts tokens on one or more incoming edges and selects outgoing edge from two or more outgoing flows. The notation for a decision node is a diamond-shaped symbol.
4.	Initial activity		Initial node is a control node at which flow starts when the activity is invoked. Activity may have more than one initial node. Initial nodes are shown as a small solid circle.
5.	Final activity		Final node is a control final node that stops all flows in an activity. Activity final nodes are shown as a solid circle with a hollow circle inside. It can be thought of as a goal notated as "bull's eye," or target.
6.	Fork		A fork in the activity diagram has a single incoming transition and multiple outgoing transitions exhibiting parallel behavior.The incoming transition triggers the parallel outgoing transitions.
7.	Join		A join in the activity diagram synchronizes the parallel behavior started at a fork. Join ascertains that all the parallel sets of activities (irrespective of the order) are completed before the next activity starts. It is a synchronization point in the diagram. Each fork in an activity diagram has a corresponding join where the parallel behavior terminates.

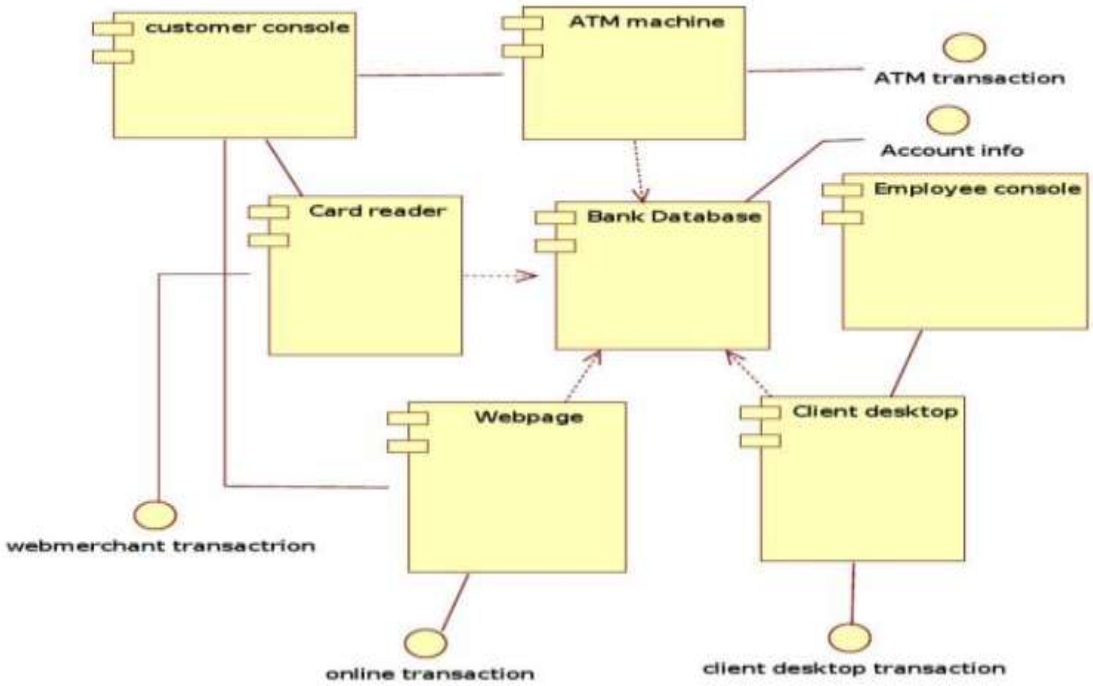

act LogIn Activity Diagram
Customer
ATM
Bank
ActivityInitial
Insert Card
Authorize Card
Ok?
[Yes]
[No]
Display Error Msg
Eject Card
Max Num Tries?
[No]
ActivityFinal
[No]
Inser Pin
Authorize PIN
PIN is correct?
[Yes]
[Yes]
Display Transactions List
ActivityFinal
Lock Card
Display Error
ActivityFinal



Step1: First initial state is created.
Step2: After that it goes to the action state insert card.

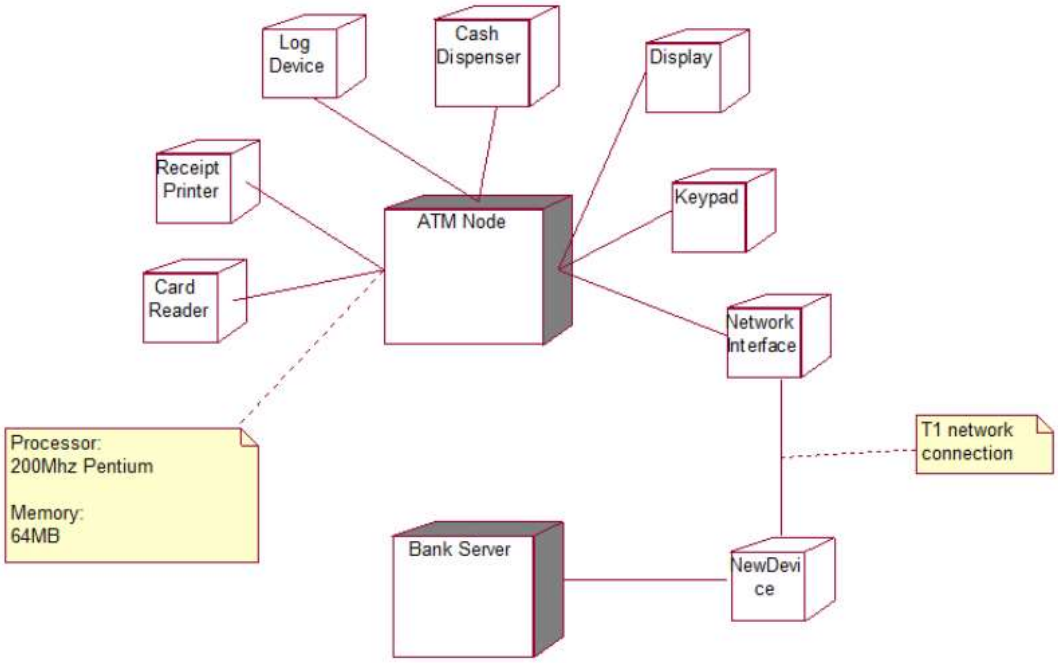
Step3: Next it undergoes transition to the state enter pin
Step4: In this way it undergoes transitions to the various states.
Step5: Use forking and joining wherever necessary.

Component diagram: Component diagrams are used to model physical aspects of a system. Physical aspects are the elements like executables, libraries, files, documents etc which resides in a node. So component diagrams are used to visualize the organization and relationships among components in a system. These diagrams are also used to make executable systems. Component diagrams can be described as a static implementation view of a system. Static implementation represents the organization of the components at a particular moment.



Deployment diagram

- Deployment diagrams are used to visualize the topology of the physical components of a system where the software components are deployed.
- So deployment diagrams are used to describe the static deployment view of a system. Deployment diagrams consist of nodes and their relationships.


Log Device
Cash Dispenser
Display
Receipt Printer
ATM Node
Keypad
Card Reader
Network Interface
Processor:
200Mhz Pentium
Memory:
64MB
T1 network connection
Bank Server
NewDevi ce